Lecture 11: Column Buckling

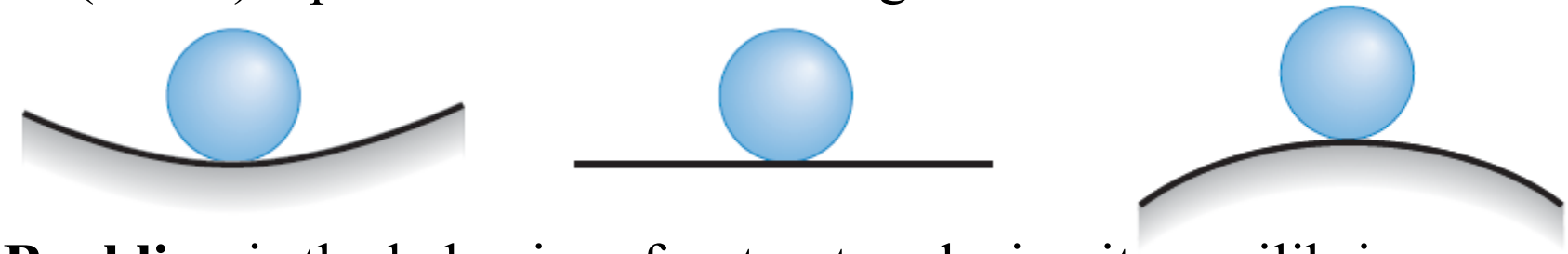
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Stability and Buckling

- **Stability** is characterized as the ability of a structure to maintain its (stable) equilibrium under working conditions.



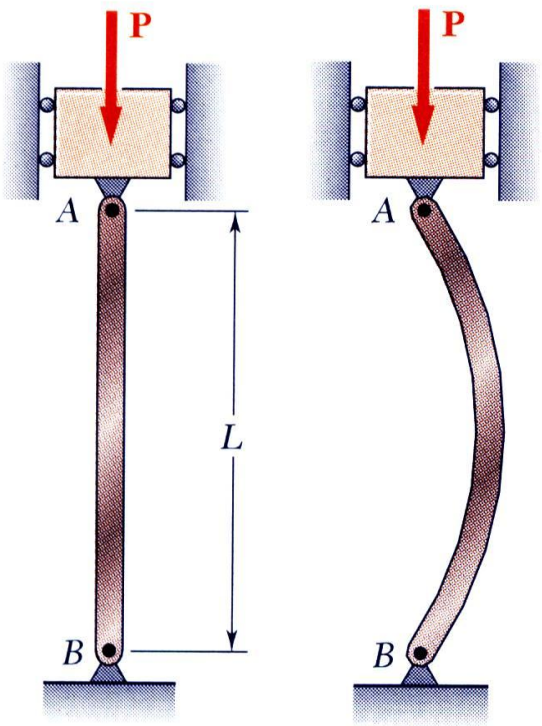
- **Buckling** is the behavior of a structure losing its equilibrium under working conditions. This is another type of failure criterion, in addition to strength (fracture/yielding), stiffness (deformation) and fatigue criteria.





- Buckling occurs suddenly and results in catastrophic accident.

Buckling and stability



Buckling of a slender column due to axial compressive load P

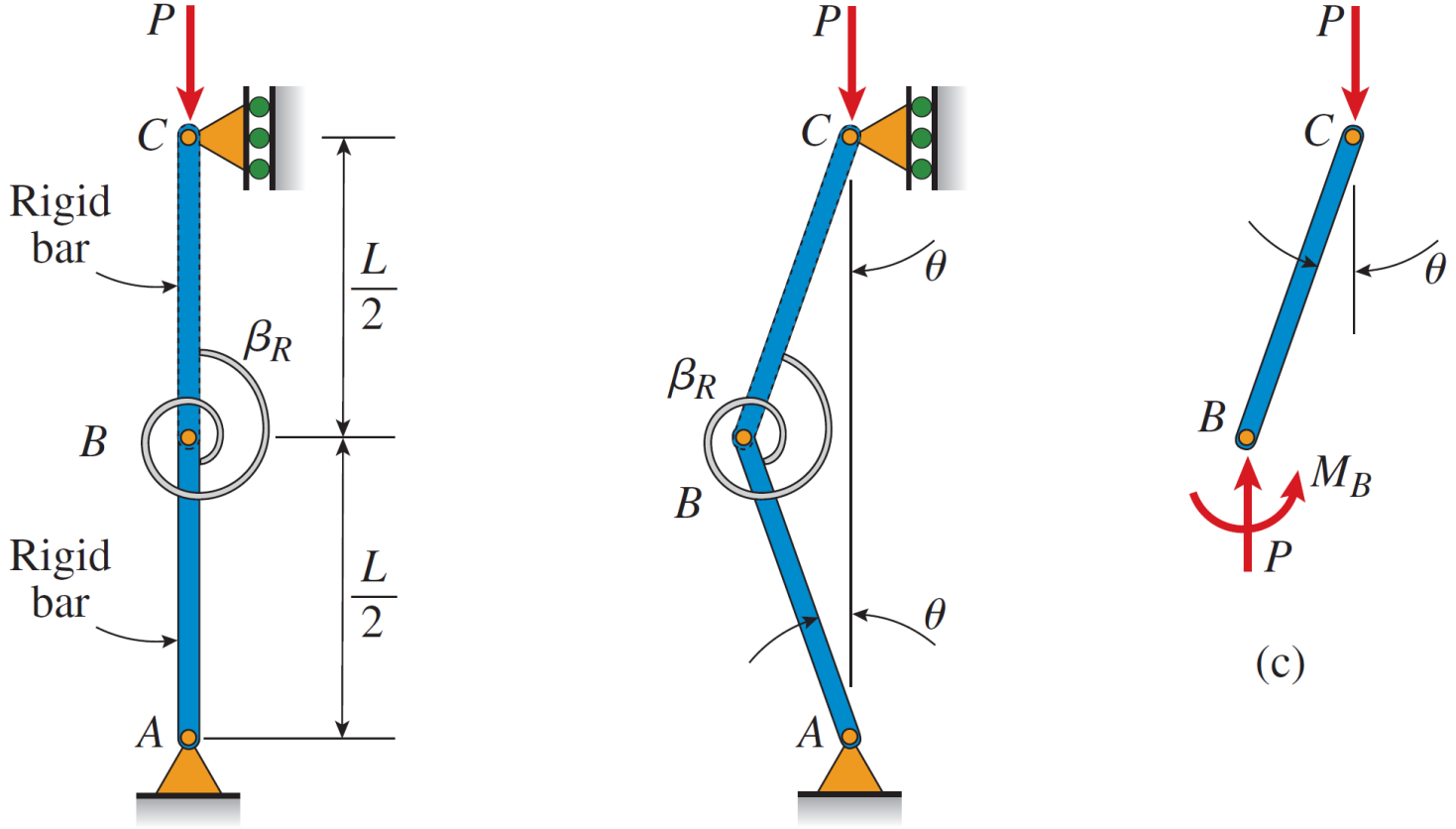
- In the conventional design of columns, cross-sectional area is selected such that maximum stress is not exceeded the allowable stress

$$\sigma = \frac{P}{A} \le [\sigma]$$

- The slender column becomes unstable suddenly subject to a critical load, **buckling** to a curved shape.

Buckling of rigid bars connected by rotational spring

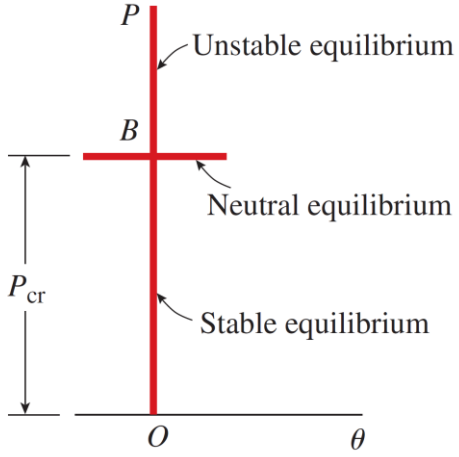
Buckling of an **idealized structure** (or buckling model) consisting of two rigid bars and a rotational spring with rotational spring constant of β_R

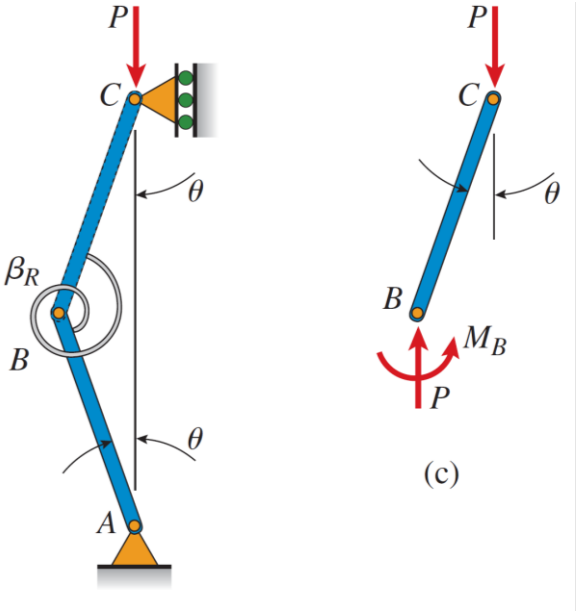


Suppose that the structure is disturbed by some external force that point *B* to move a small distance laterally. The rotational spring at B provides **restoring moment** and tendency of P is to increase lateral displacement.

Buckling and stability of rigid bars connected by spring

- If P is very small, the action of restoring moment will dominate and the structure will return to initial straight position: **Stable.**
- If P is huge, the structure will collapse. The structure is **unstable** and fails by **lateral buckling.** The transition between **stable** and **unstable** condition happen at axial force known as **critical load**.
- Consider bar *BC* as a free body and note that it is subjected to the action of the axial forces *P* and moment M_B in the spring.





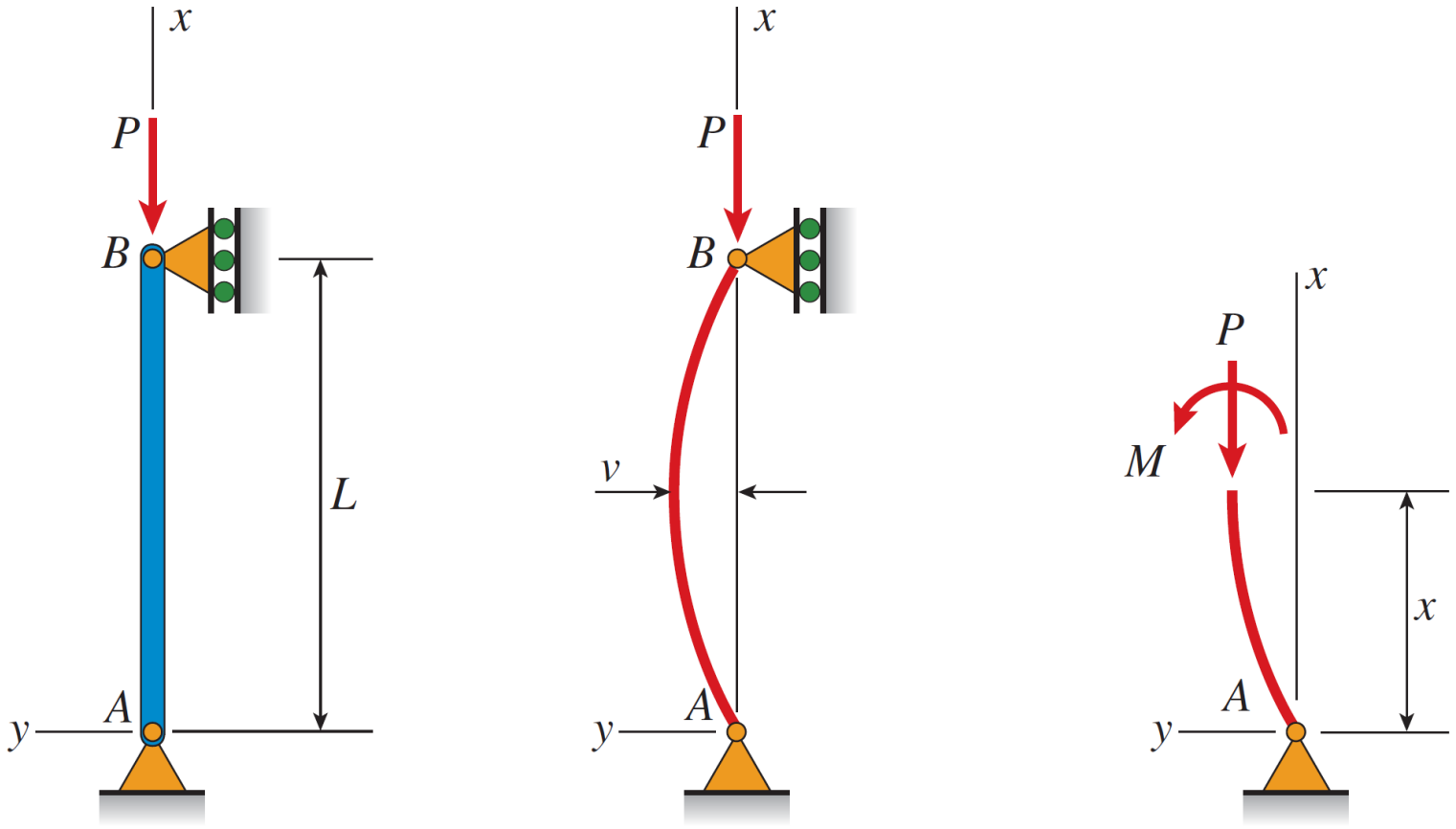
$$M_B = 2\beta_R\theta \qquad M_B - P\left(\frac{\theta L}{2}\right) = 0$$

$$\left(2\beta_R - \frac{PL}{2}\right)\theta = 0 \qquad P_{cr} = \frac{4\beta_R}{L}$$

If $P < P_{cr}$, the structure is *stable*.
If $P > P_{cr}$, the structure is *unstable*.

Buckling of columns with pinned ends

Ideal column: Analyzing a slender column with pinned ends loaded by a vertical force P applied through the centroid of the end section. The column is perfectly straight and made of linear elastic material.

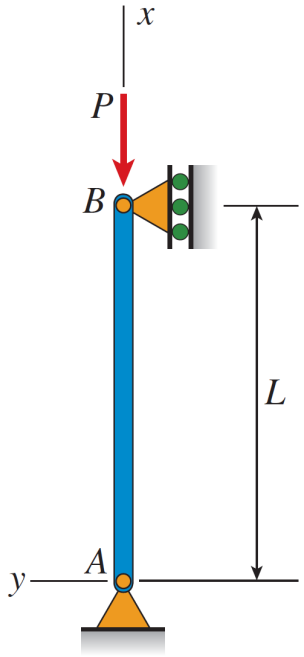


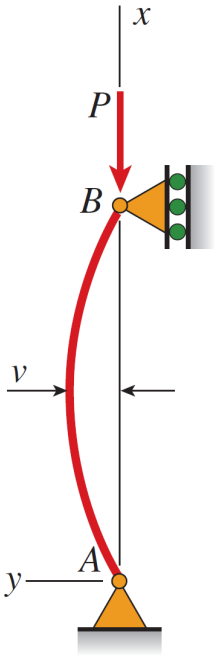
Column Buckling of pinned column

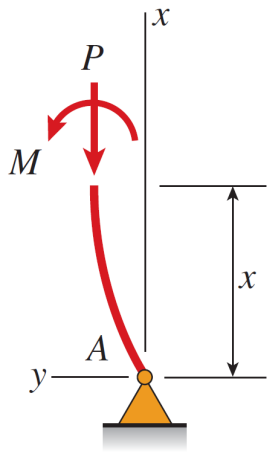
Differential equations of the deflection curve of a beam. M is the bending moment at any cross section, v is the lateral deflection in the y direction and EI is the flexural rigidity for bending in the xy plane.

$$EIv'' = M$$

- Note: second-order derivative of deflection v'' is curvature κ







- Moment Equilibrium of the free body w.r.t A hinge:

$$M + Pv = 0$$

$$M = -Pv$$

Column Buckling of pinned column

Differential equations of the deflection curve is obtained below:

$$EIv'' = M \qquad M = -Pv$$

$$EIv'' + Pv = 0$$

By solving this *homogeneous, linear, differential equation of second order with constant coefficients*, we can determine the critical load and deflected shape of the buckled column.

We introduce the notation k to solve it:

$$k^2 = \frac{P}{EI} \quad \text{or} \quad k = \sqrt{\frac{P}{EI}} \qquad v'' + k^2 v = 0$$

From mathematics we know that the **general solution** of this equation

$$v = C_1 \sin kx + C_2 \cos kx$$

where C_1 and C_2 are constants of integration to be evaluated from boundary conditions

Column Buckling of pinned column

To evaluate the constants of integration, we use the boundary conditions at the ends of the column, the deflections are 0 at both ends:

$$v = C_1 \sin kx + C_2 \cos kx$$

$$v(0) = 0 \quad \text{and} \quad v(L) = 0$$

The first condition gives C_2=0 and therefore:

$$v = C_1 \sin kx \qquad C_1 \sin kL = 0$$

If the constant C1 is 0, the deflection is 0 and not our target:

$$\sin kL = 0 \qquad kL = n\pi \qquad n = 1, 2, 3, \ldots$$

$$P = \frac{n^2\pi^2 EI}{L^2} \qquad n = 1, 2, 3, \ldots$$

$$v = C_1 \sin kx = C_1 \sin \frac{n\pi x}{L} \qquad n = 1, 2, 3, \ldots$$

Column Buckling of pinned column

The lowest critical load for a column with pinned ends is obtained when n=1:

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

The corresponding buckled shape (mode shape) is

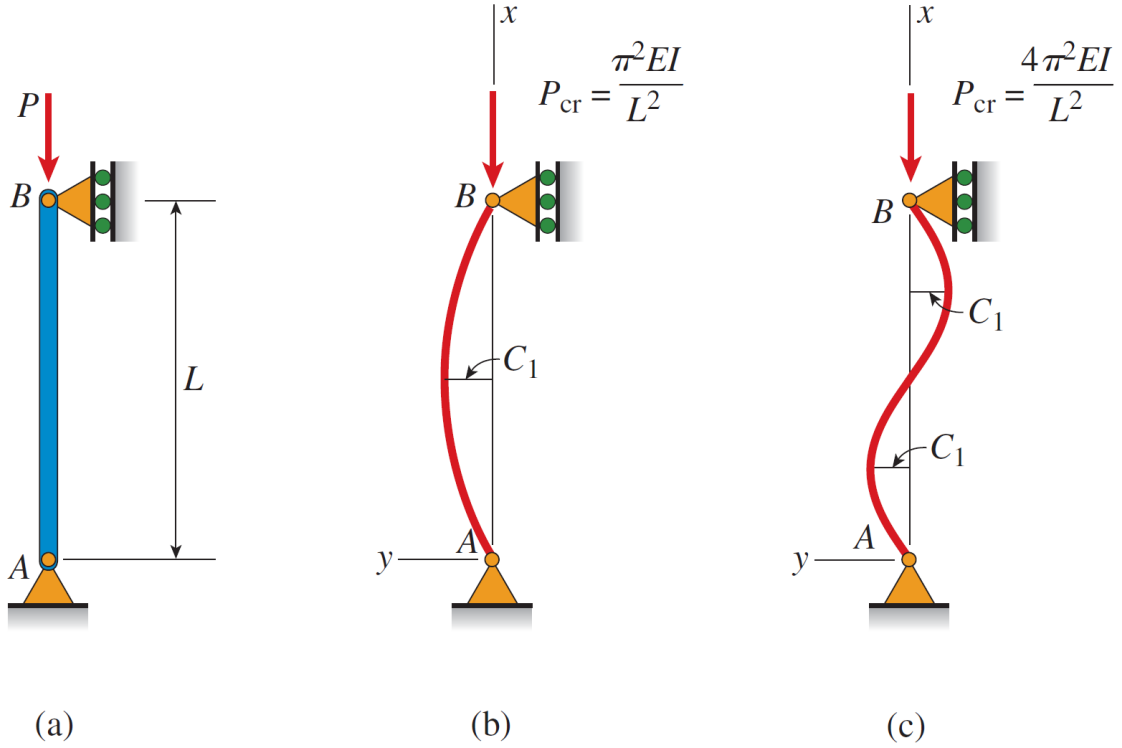
$$v = C_1 \sin \frac{\pi x}{L}$$

constant C_1 is the deflection at midpoint of the column and may have any small value. The first mode is called the **fundamental case of column buckling**.

This type of buckling of ideal elastic column is called **Euler Buckling** and **Euler load**.

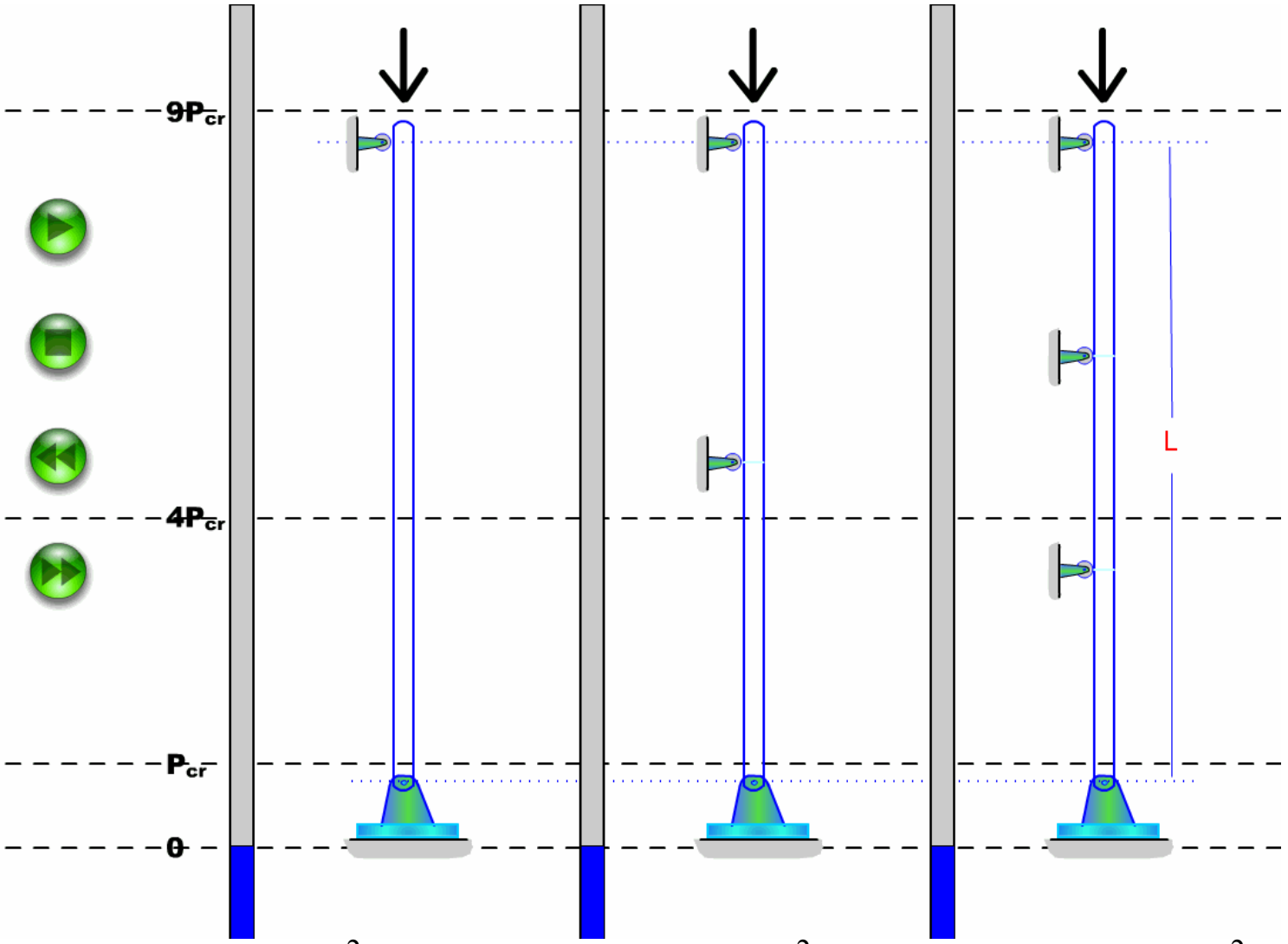
Column Buckling of pinned column

Buckled shapes for an ideal column with pinned ends: Buckled shapes for the **higher modes** are often of no practical interest because the column buckles when the axial load P reaches its lowest critical value. The only way to obtain higher modes of buckling is to provide lateral support of the column at intermediate points



Buckling shape of n=1 Buckling shape of n=2

Buckling Modes



$$P_{cr} = \frac{EI\pi^2}{L^2};$$

$$P_{cr} = \frac{4EI\pi^2}{L^2};$$

$$P_{cr} = \frac{9EI\pi^2}{L^2}$$

Critical Stress

We can calculate the corresponding **critical stress** by dividing the critical buckling load by the cross-sectional area A. For the fundamental case of buckling the critical stress is

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 EI}{AL^2}$$

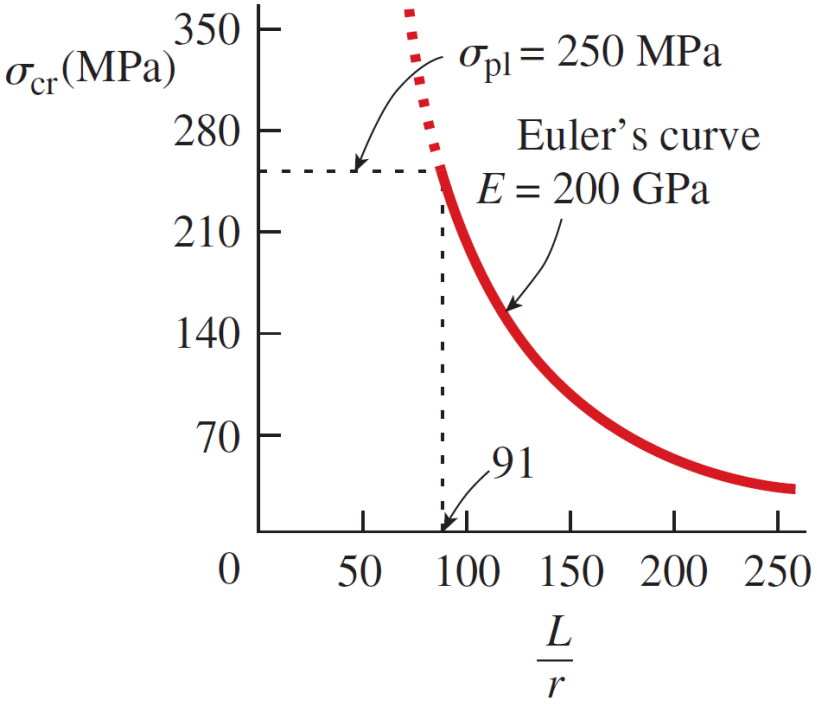
in which I is the moment of inertia for the principal axis about which buckling occurs. This equation can be written in a more useful form by introducing the notation, where r is the **radius of gyration** of the cross section in the plane of bending.

$$r = \sqrt{\frac{I}{A}} \qquad \sigma_{cr} = \frac{\pi^2 E}{(L/r)^2}$$

in which $\lambda = L/r$ is a nondimensional ratio called the **slenderness ratio**

Critical Stress and Euler's curve

We can plot a graph of the stress as a function of the slenderness ratio and obtain **Euler's curve.**

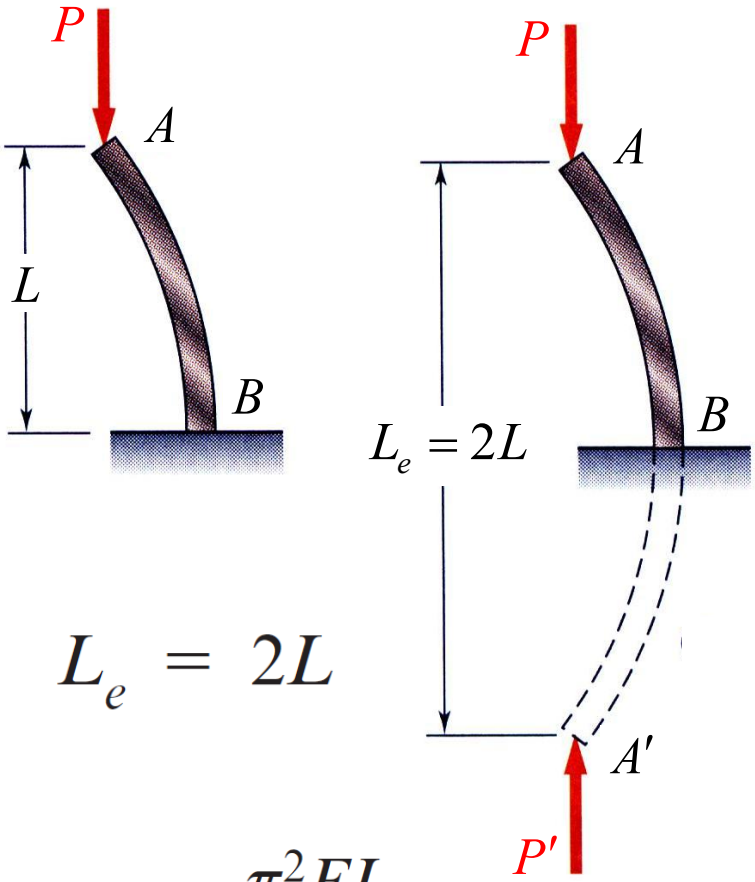


$$\sigma_{cr} = \frac{\pi^2 E}{(L/r)^2}$$

in which $\lambda = L/r$ is a nondimensional ratio called the **slenderness ratio**

The curve is valid only when the critical stress is less than the proportional limit of the steel. Therefore, we draw a horizontal line on the graph at the proportional limit of the steel (assumed to be 250 MPa) and terminate Euler's curve at that level of stress

Cantilevered Columns



$$L_e = 2L$$

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

- A column with one fixed and one free end, will behave as the upper-half of a pin-connected column.
- The effective length L_e for any column is the length of the equivalent pinned-end column.
- L_e is the length of a pinned-end column having a deflection curve that exactly matches all or part of the deflection curve of the original column.
- L_e is the distance between pints of inflection (that is points of zero moment) in the deflection curve.

Effective Length

- The effective length L_e is often expressed in terms of an **effective length factor *K*** (or μ)

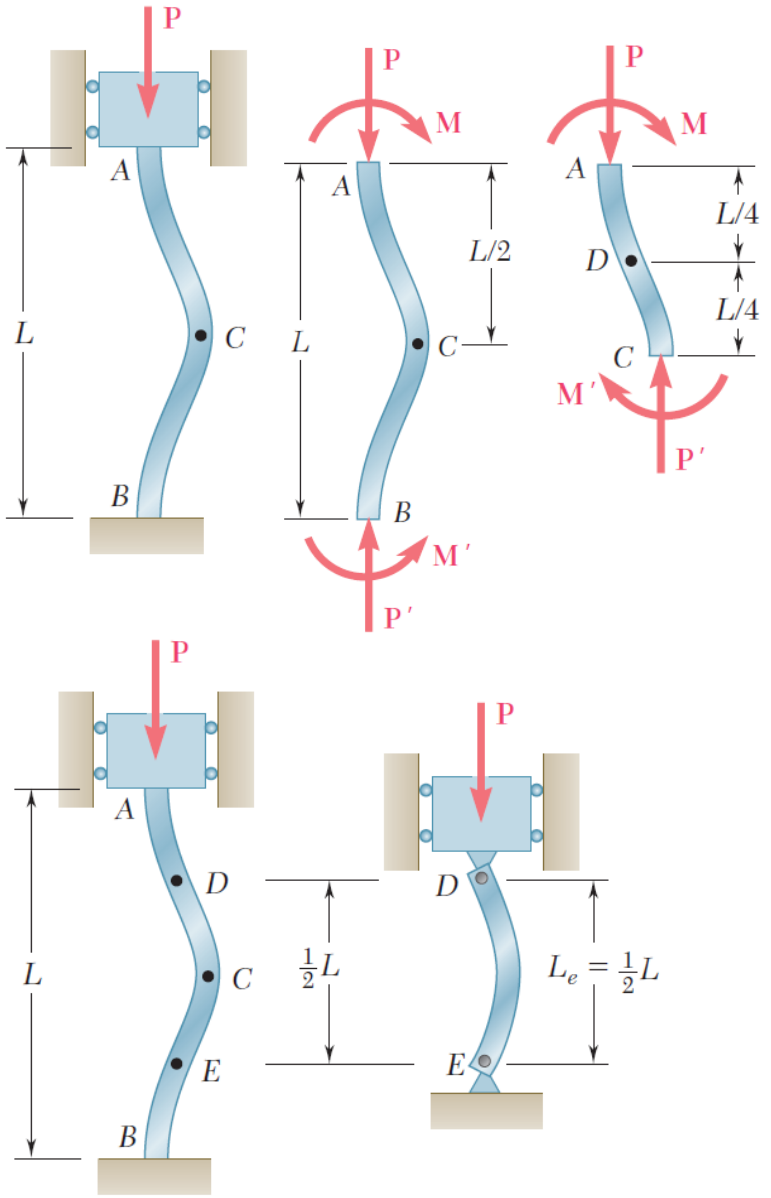
$$L_e = KL$$

where L is the actual length of the column. Thus, the critical load is

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

The factor K is 2 for a column fixed at base and free at top and is 1 for pinned-end column.

Columns with Two Fixed Ends

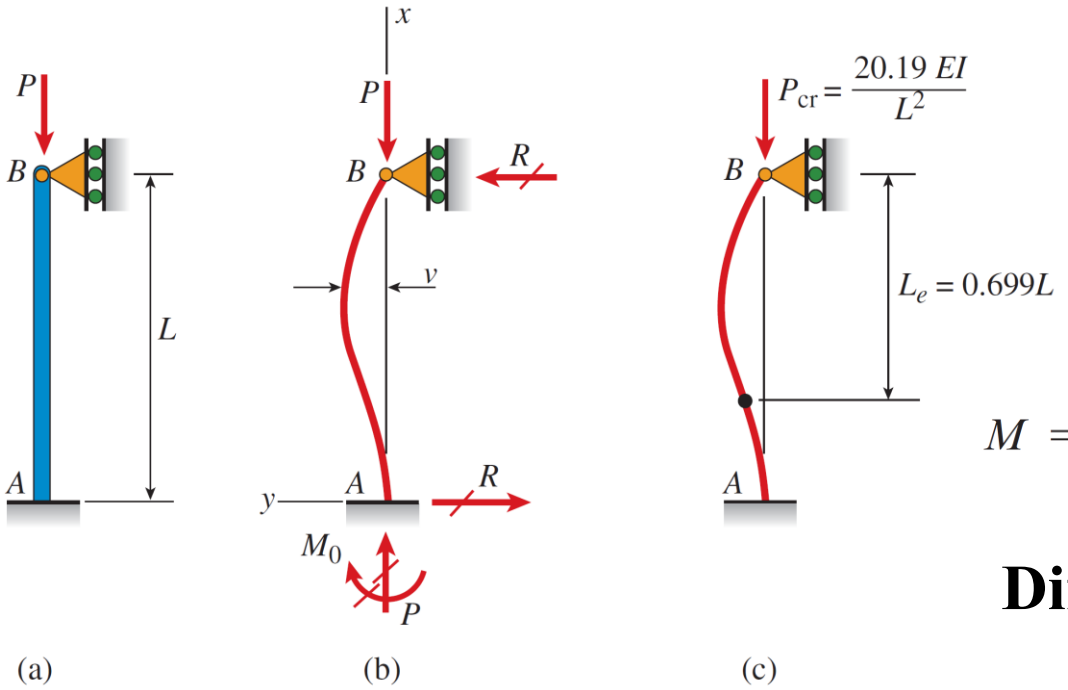


- The symmetry of the supports and of the loading requires that the shear at *C* and the horizontal reactions at both ends be zero.
- The equation of the deflection curve involves sine and cosine functions.
- Point *D* must be a point of inflection, where the bending moment is zero.
- It follows that the portion *DE* of the column must behave as a pin ended column.

$$L_e = 0.5L \quad \Rightarrow P_{cr} = \frac{\pi^2 EI}{L_e^2} = \frac{4\pi^2 EI}{L^2}$$

Columns with One Fixed End and One Pinned End

When the column buckles, a reactive moment M_0 develops at the base because there can be no rotation at that point. Then from the equilibrium of the entire column, we know that there must be horizontal reactions R at each end such that $M_0 = RL$



The bending moment in the buckled column, at distance x from the base is

$$M = M_0 - Pv - Rx = -Pv + R(L - x)$$

Differential equation is

$$EIv'' = M = -Pv + R(L - x)$$

Columns with One Fixed End and One Pinned End

- Substituting $k^2=P/EI$ we get

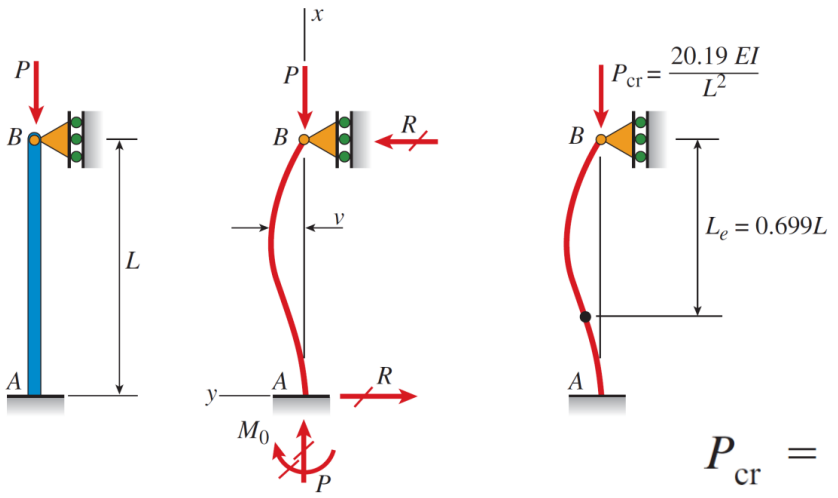
$$v'' + k^2 v = \frac{R}{EI}(L - x)$$

- The general solution of this equation is

$$v = C_1 \sin kx + C_2 \cos kx + \frac{R}{P}(L - x)$$

We need three **boundary conditions** to solve coefficients:

$$v(0) = 0 \qquad v'(0) = 0 \qquad v(L) = 0$$



$$C_2 + \frac{RL}{P} = 0 \qquad C_1 k - \frac{R}{P} = 0$$

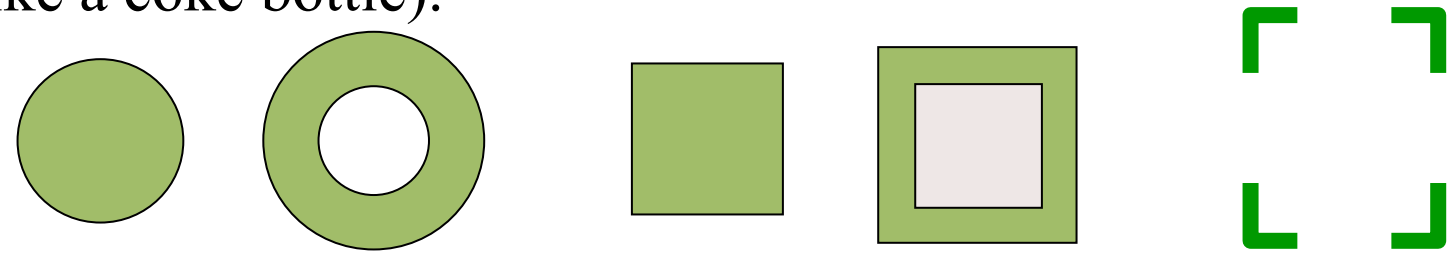
$$C_1 \tan kL + C_2 = 0$$

$$kL = \tan kL \quad \Rightarrow \quad kL = 4.4934$$

$$P_{cr} = \frac{20.19EI}{L^2} = \frac{2.046\pi^2 EI}{L^2} \qquad L_e = 0.699L \approx 0.7L$$

Ways to Improve Column Stability

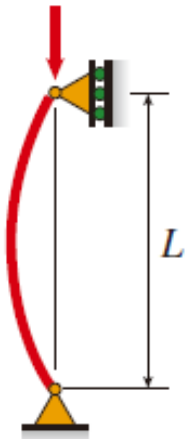
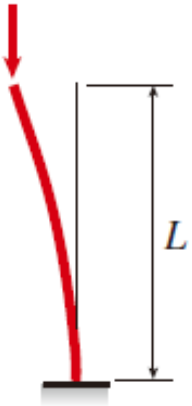
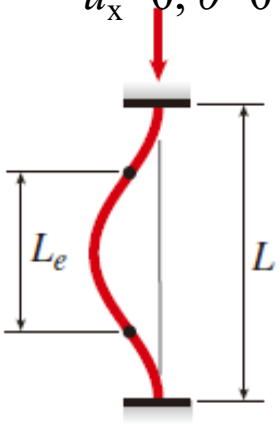
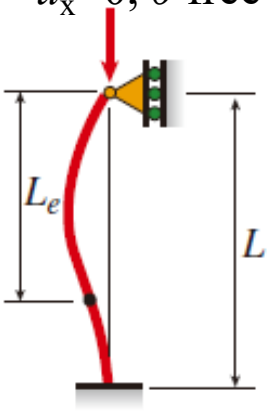
- Selection of materials: higher elastic modulus.
- Decrease effective column length (*KL*).
- Increase moment of inertia for a given cross-sectional area, such as increasing the radius and reduce the thickness of thin-walled section (but if thickness is too small, there may be local buckling, like a coke bottle).



- Coordination of end conditions and moment of inertia in orthogonal planes
- For a group of columns, make each one equally stable

Ways to Improve Column Stability

- Decrease effective column length (KL) has notable effect on buckling load

(a) Pinned-pinned column	(b) Fixed-free column	(c) Fixed-fixed column	(d) Fixed-pinned column
$P_{cr} = \frac{\pi^2 EI}{L^2}$	$P_{cr} = \frac{\pi^2 EI}{4L^2}$	$P_{cr} = \frac{4\pi^2 EI}{L^2}$	$P_{cr} = \frac{2.046\ \pi^2 EI}{L^2}$
u_x=0, θ free  u_x=u_y=0, θ free	u_x and θ both free  u_x=u_y=0, θ=0	u_x=0, θ=0  u_x=u_y=0, θ=0	u_x=0, θ free  u_x=u_y=0, θ=0
$L_e = L$	$L_e = 2L$	$L_e = 0.5L$	$L_e = 0.699L$
$K = 1$	$K = 2$	$K = 0.5$	$K = 0.699$

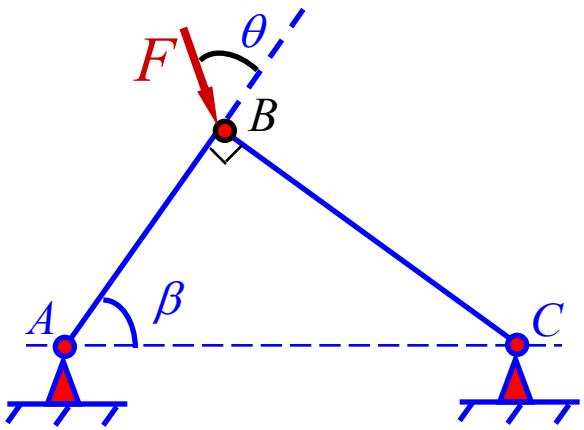
Example 9.1 Truss buckling

For the truss shown, F, β, and L_{AC} are given. Find θ under which column AB and BC reach critical stability simultaneously. Neglect out-of-plane buckling.

- Solution:

1. Euler's equation for column AB:

$$F_{AB} = F\cos\theta = \frac{\pi^2 EI}{l_{AB}^2} = \frac{\pi^2 EI}{l^2\cos^2\beta}$$



2. Euler's equation for column BC:

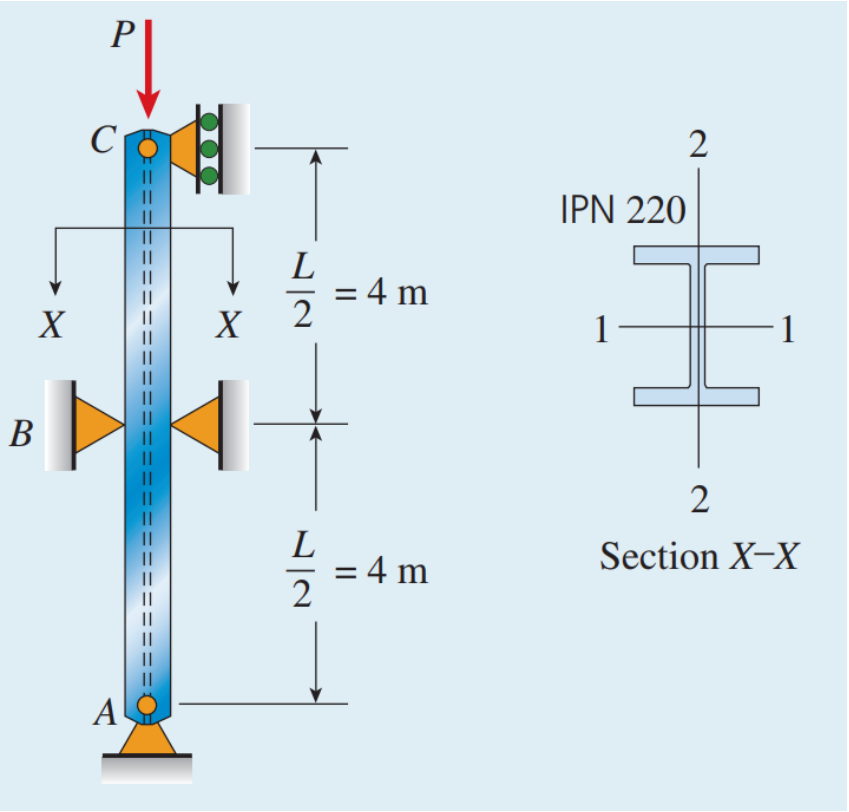
$$F_{BC} = F\sin\theta = \frac{\pi^2 EI}{l_{CB}^2} = \frac{\pi^2 EI}{l^2\sin^2\beta}$$

3. If column AB & BC reach critical stability simultaneously:

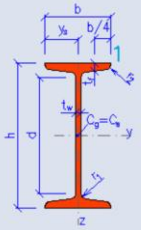
$$\left.\begin{aligned} F\cos\theta &= \frac{\pi^2 EI}{l^2\cos^2\beta} \\ F\sin\theta &= \frac{\pi^2 EI}{l^2\sin^2\beta} \end{aligned}\right\} \Rightarrow \cot\theta = \tan^2\beta$$

Example 9.2: Column with different constraints

A long, slender column ABC is pin supported at the ends and compressed by an axial load P. Lateral support is provided at the midpoint B in the plane of the figure. However, lateral support perpendicular to the plane of the figure is provided only at the ends. The column is constructed of a standard steel shape (IPN 220) having modulus of elasticity E=200GPa and proportional limit σ_{pl}=300MPa. The total length of the column is 8m. Determine the allowable load P_{allow} using a factor of safety with respect to Euler buckling of the column. Neglect self-weight.

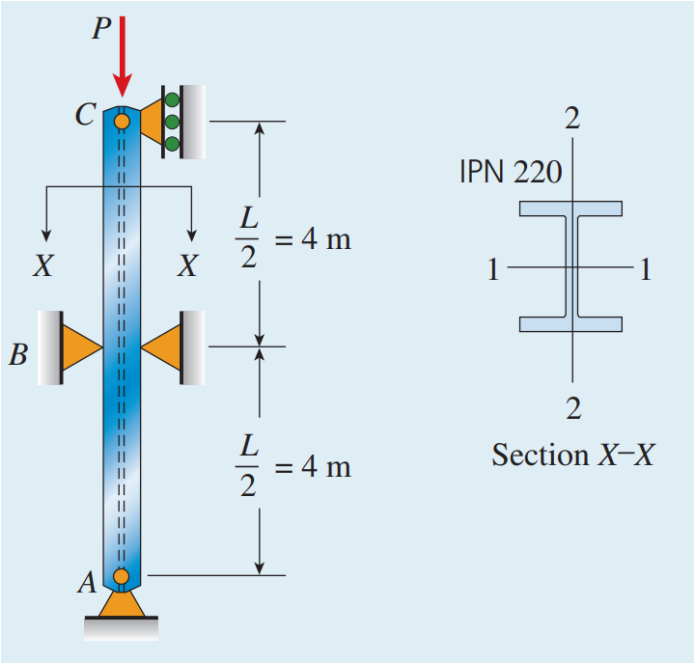


[Section properties - Section IPN 220 (staticstools.eu)]

IPN 220			
Geometry		**Section properties**	
h = 220 mm		Axis y	Axis z
b = 98 mm		I_y = 3.05E+7 mm^4	I_z = 1.62E+6 mm^4
t_f = 12.2 mm		W_{y1} = 2.78E+5 mm^3	W_{z1} = 3.31E+4 mm^3
t_w = 8.1 mm		$W_{y,pl}$ = 3.22E+5 mm^3	$W_{z,pl}$ = 5.46E+4 mm^3
r_1 = 8.1 mm		i_y = 87.9 mm	i_z = 20.3 mm
r_2 = 4.9 mm		S_y = 1.61E+5 mm^3	S_z = 2.73E+4 mm^3
y_s = 49 mm		**Warping and buckling**	
d = 175.8 mm	G = 31 kg.m^{-1}	I_w = 1.69E+10 mm^6	I_t = 1.87E+5 mm^4
A_L = 0.77 m^2.m^{-1}	A = 3950 mm^2	i_w = 22.9 mm	i_{pc} = 90.2 mm

This column has strong axis and weak axis.
Strong axis radius=87.9mm
Weak axis radius=20.3mm

Example 9.2: Column with different constraints



In this problem, buckling in the plane of this figure is weak axis (rotating with respect to 2-2 axis). Need to use I_2 in Euler's equation with effective length of 4m.

Buckling out-of-plane is rotating w.r.t. strong axis 1-1, and needs to use I_1 in Euler's equation.

$I_1 = 3060\ \text{cm}^4$ $I_2 = 162\ \text{cm}^4$ $A = 39.5\ \text{cm}^2$

Critical loads. If the column buckles in the plane of the figure, the critical load is

$$P_{cr} = \frac{\pi^2 E I_2}{(L/2)^2} = \frac{4\pi^2 E I_2}{L^2}$$

Substituting numerical values, we obtain

$$P_{cr} = \frac{4\pi^2 E I_2}{L^2} = \frac{4\pi^2(200\ \text{GPa})(162\ \text{cm}^4)}{(8\ \text{m})^2} = 200\ \text{kN}$$

If the column buckles perpendicular to the plane of the figure, the critical load is

$$P_{cr} = \frac{\pi^2 E I_1}{L^2} = \frac{\pi^2(200\ \text{GPa})(3060\ \text{cm}^4)}{(8\ \text{m})^2} = 943.8\ \text{kN}$$

Therefore, the critical load for the column (the smaller of the two preceding values) is

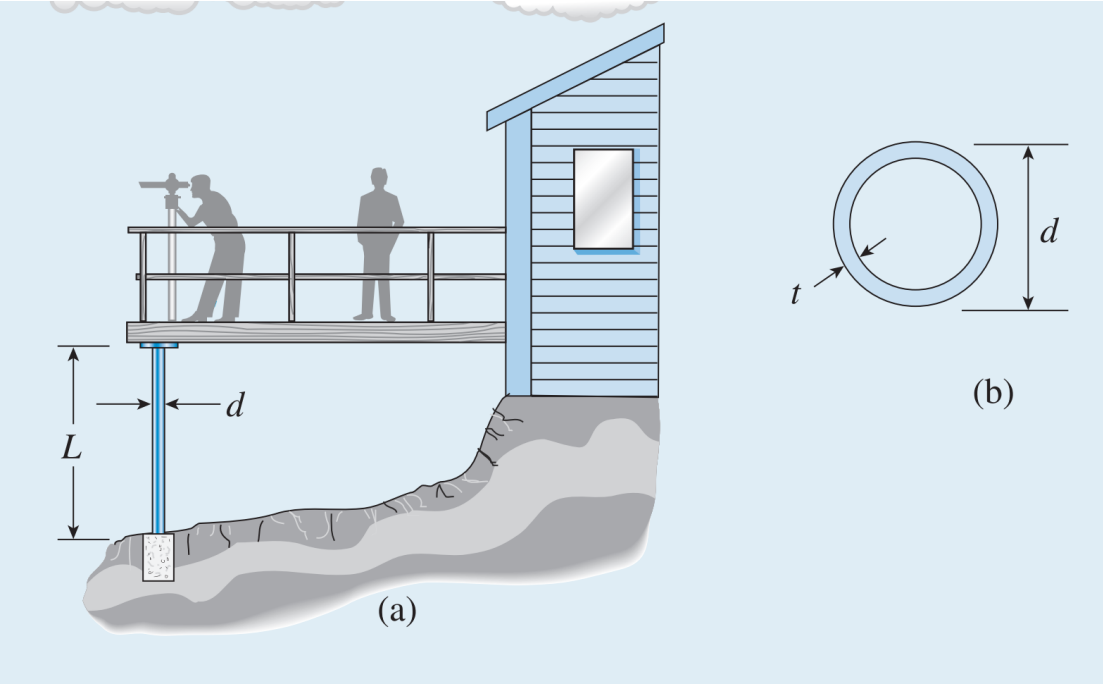
$$P_{cr} = 200\ \text{kN}$$

$$P_{\text{allow}} = \frac{P_{cr}}{n} = \frac{200\ \text{kN}}{2.5} = 79.9\ \text{kN}$$

The stress due to 200 kN is 50 MPa, lower than the material strength (proportional limit).

Example 9.3: A pipe under compression

A viewing platform in a wild-animal park is supported by a row of aluminum pipe columns having length L=3.25m and outer diameter d=100mm. The bases of the columns are set in concrete footings and the tops of the columns are supported laterally by the platform. The columns are being designed to support compressive loads P=100kN. Determine the minimum required thickness t of the columns if a factor of safety is required with respect to Euler buckling. (For the aluminum, use 72 GPa for the modulus of elasticity and use 480 MPa for the proportional limit.)

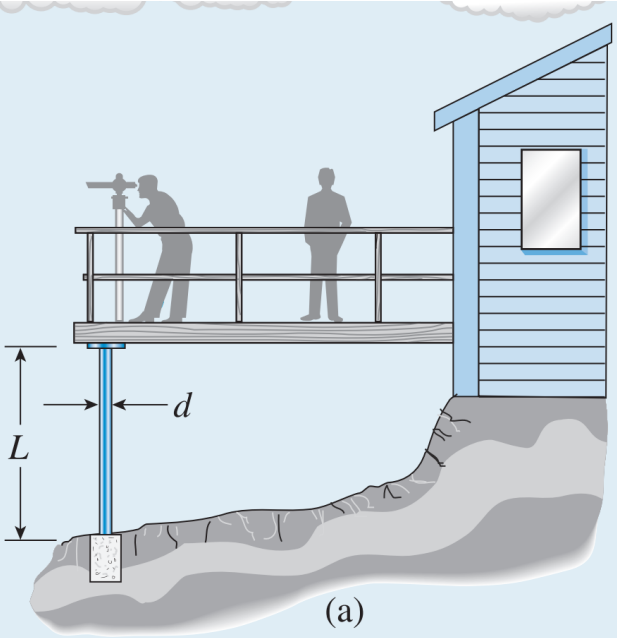


Check the constraint at both ends: the pipe bottom is cast into concrete; this is a fixed connection.

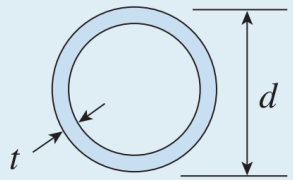
At top, the lateral displacement of the column is restricted. This is because the axial displacement of roof can be assumed small parameter. However, the rotation of the column top is permitted. (when column buckle, the roof can have rotation angle, not negligible)

This is equivalent to fixed-pinned condition. $L_{eff}=0.7L$

Example 9.3: A pipe under compression



(a)



(b)

$$P_{cr} = \frac{2.046\pi^2 EI}{L^2}$$

$$I = \frac{\pi}{64}[d^4 - (d - 2t)^4]$$

$$I = \frac{\pi}{64}[(0.1\text{ m})^4 - (0.1\text{ m} - 2t)^4]$$

$$P_{cr} = nP = 3(100\text{ kN}) = 300\text{ kN}$$

$$300{,}000\text{ N} = \frac{2.046\,\pi^2(72 \times 10^9\text{ Pa})}{(3.25\text{ m})^2}\left(\frac{\pi}{64}\right)[(0.1\text{ m})^4 - (0.1\text{ m} - 2t)^4]$$

$$t_{min} = 6.83\text{ mm}$$

$$I = \frac{\pi}{64}[d^4 - (d - 2t)^4] = 2.18 \times 10^6\text{ mm}^4$$

$$A = \frac{\pi}{4}[d^2 - (d - 2t)^2] = 1999\text{ mm}^2 \qquad r = \sqrt{\frac{I}{A}} = 33.0\text{ mm}$$

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{300\text{ kN}}{1999\text{ mm}^2} = 150\text{ MPa}$$

Summary

- Buckling instability of slender columns is an important mode of failure that must be considered in their design (in addition to strength and stiffness).
- A slender column with pinned ends and length L, acted on by a compressive load at the centroid of the cross section, and restricted to linear elastic behavior, will buckle at the **Euler buckling load**

$$P_{cr} = \pi^2 EI/L^2$$

- Changing the support conditions, or providing additional lateral supports, changes the critical buckling load.
- P_{cr} for these **other support cases** may be obtained by replacing the actual column length (L) by the **effective length** (L_e).

$$L_e = KL \qquad P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$